



Analytics Solutions Bootcamp: #01

Foundations of Analytics Solutions

Learn what analytics solutions are and how they solve business problems.

Topics to be covered



- What is an analytics solution?
- Analytics solutions across data practices
- Real-world analytics solutions examples across industries
- Approach to designing and building analytics solutions
 - INFORMS Analytics Framework
- Introduction to data careers and career progression
 - AAP Philippine Skills Framework

What is an analytics solution?

Definition of analytics solution >>

Analytics is the process of collecting and analyzing data to discover, interpret, and communicate meaningful insights.

A **solution** is a combination of products, services, processes, and technologies designed to address a specific customer problem or meet a strategic business need.

What is an analytics solution?

Analytics Solution in a nutshell >>

Analytics Solution aims to address business problems through data analytics.

1. We have a **clear** business problem.
2. We have a **clear baseline measure** of how we can quantify the problem.
3. We have **data** to analyze and explore.

Types of analytics solutions

Analytics Solutions Projects



Domain	Solution
Data Engineering	Data ingestion pipelines (ETL)
	Database migration projects
	Data warehouse / data lake implementation
Data Analysis / Business Intelligence	Executive reports and dashboards
	Exploratory data analysis
	Business domain experts

Domain	Solution
Data Science	Forecasting model
	Classification model
	Recommendation engine
Data Governance	Customer Unification (C360)
	Master Data Management
	Data cataloging

Real-world analytics solutions examples across industries

Real-world industry examples



Industry	Business Problem	Analytics Solution
Human Resources	Limited visibility into workforce composition and trends	Employee demographics dashboard
	High employee turnover	Employee attrition analysis
Retail	Understanding customer behavior	Customer segmentation
	Forecasting demand	Sales forecasting model
Banking / FinTech	Customers dropping out of account opening process	Funnel analysis
	High customer churn	Customer churn analysis
	Fraudulent transactions	Fraud detection and alerting solution

Real-world analytics solutions examples across industries

Real-world industry examples



Industry	Business Problem	Analytics Solution
Real Estate	Difficulty tracking financial performance	Financial reporting dashboard (AR, budget vs actual, cash flow)
	Low occupancy rates	Occupancy and tenant analytics dashboard
	Uncertainty in property pricing	Property valuation model
E-Commerce	Low conversion rates	Website funnel analysis
	Poor product recommendations	Recommendation engine
	High cart abandonment	Cart abandonment analysis
Healthcare	Long patient waiting times	Patient flow analytics
	Resource allocation challenges	Hospital utilization dashboard

Real-world analytics solutions examples across industries

Real-world industry examples



Industry	Business Problem	Analytics Solution
Manufacturing	Unexpected equipment failures	Predictive maintenance solution
	Production bottlenecks	Production efficiency dashboard
	Product quality issues	Quality control analytics
Logistics & Supply Chain	Late deliveries	Delivery performance dashboard
	Inefficient routes	Route optimization analytics
	Inventory imbalances	Supply chain visibility dashboard
Telecommunications	Customer attrition	Subscriber churn prediction
	Network disruptions	Network monitoring analytics
	Service quality concerns	Customer experience dashboard

Real-world analytics solutions examples across industries

Real-world industry examples



Industry	Business Problem	Analytics Solution
Education	Student dropouts	Student retention analysis
	Low academic performance	Academic performance dashboard
	Resource planning challenges	Enrollment forecasting
Marketing	Ineffective campaigns	Marketing campaign performance dashboard
	Poor lead conversion	Lead funnel analytics
	Unclear return on investment	Marketing ROI analysis
Government / Public Sector	Slow service delivery	Public service performance dashboard
	Resource allocation challenges	Budget utilization analytics
	Lack of citizen insights	Citizen feedback analysis

Notable data products



Recommendation engine
“recommends” contents for you.
It is an application of machine learning which can be embedded in apps, like Facebook, YouTube, Netflix, and Amazon.

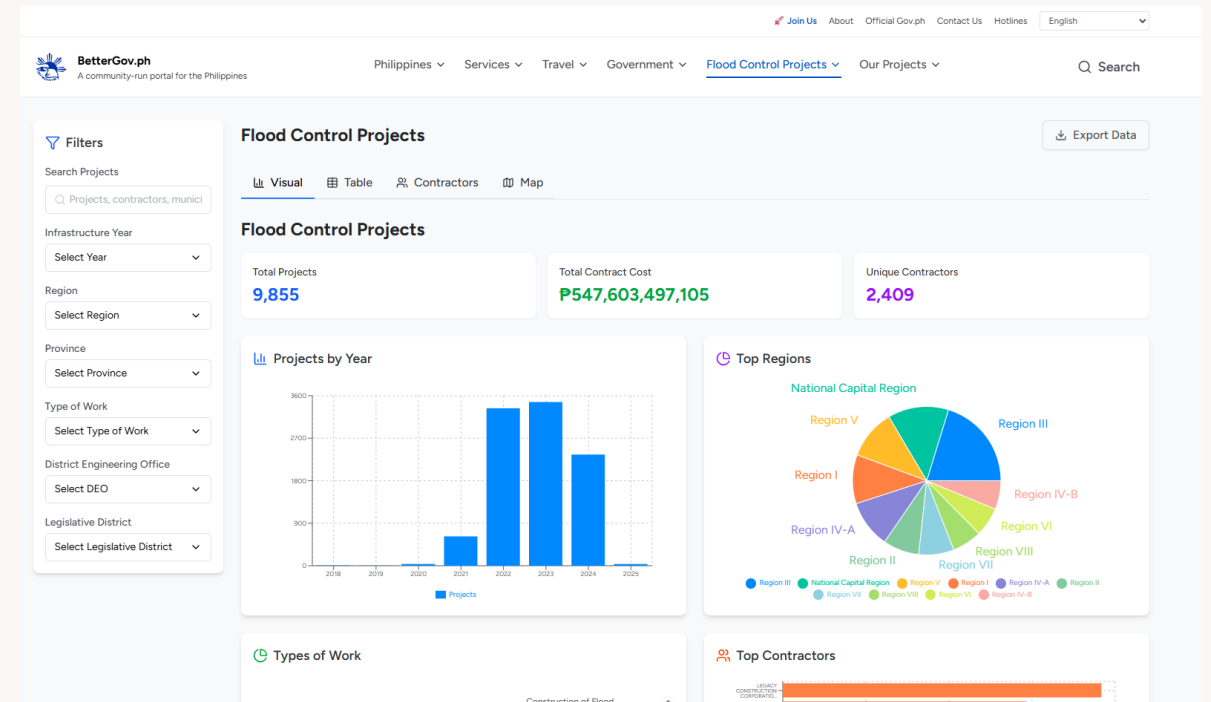


Notable data products



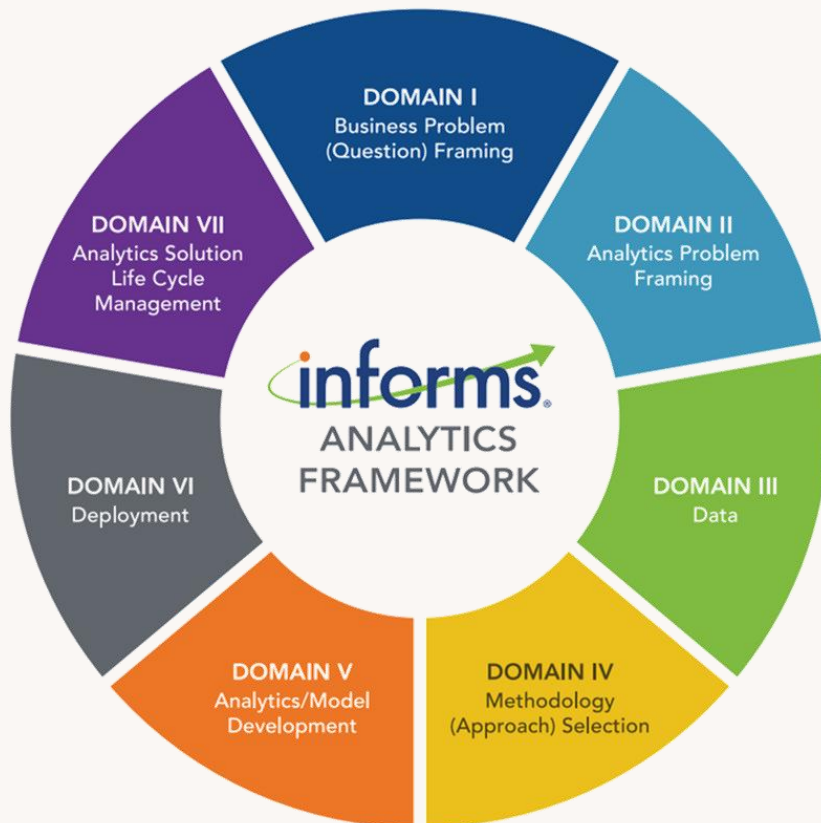
BetterGov.ph

The volunteer run portal of the Republic of the Philippines. Find information, access government services, stay updated with the latest news about the Philippines.



Framework for designing and building analytics solutions

INFORMS Analytics Framework >>



1. Business Problem Framing
2. Analytics Problem Framing
3. Data
4. Methodology Selection
5. Analytics/Model Development
6. Deployment
7. Analytics Solution Lifecycle Management

1. Business problem framing >>

Around 2023, I always ask myself “*saan napupunta pera ko?*”. I didn’t know how to answer the question, my savings are not growing, and sometimes I use my savings for my expenses.

Business problem:

- How do I save more?
- How do I monitor my finances?

2. Analytics problem framing >>

The problem is descriptive in nature. Analytics questions that need to be answered are:

- What are my largest expense categories?
- How much do I spend each month?
- Which expenses are fixed, variable?
- Which months had negative cash flow?
- How much of my income went to savings, debt, household expenses, and family support?

3. Data



The solution would require the following data:

- Income: date, amount, source
- Expenses: data, amount
- Transaction classification: category, subcategory
- Accounts: savings, e-wallet, credit card

4. Methodology Selection



The project will use descriptive analytics to organize and summarize my financial transactions. The tracker will help answer:

- How much money do I receive each month?
- How much do I spend?
- Which expense categories account for most of my spending?
- Am I spending more than my planned budget?

5. Analytics Solution Development >>

The main dataset will contain one row for every financial transaction.

- Date
- Transaction Type
- Category / Subcategory
- Account
- Amount

Main metrics to be used are as follows:

- $\text{Net Cash Flow} = \text{Total Income} - \text{Expense}$
- $\text{Budget Variance} = \text{Actual Expense} - \text{Budgeted Expense}$

6. Deployment



The completed tracker will be incorporated into my regular financial routine.

- Record each transaction immediately. Reconcile immediately.
- Categorize every income and expense transaction.
- Review category spending against the monthly budget.
- Conduct a complete financial review at the end of every month.
- Identify categories in which I exceeded my budget.

Framework for designing and building analytics solutions

7. Lifecycle Management



Daily activities: Track all cash flows. Reconcile if needed.

Weekly activities: Weekly reconciliation and estimate budget and spends.

AAP Philippine Skills Framework >>

Career Maps

Analytics & Artificial Intelligence Career Map

The Career Map guides individuals in exploring career pathways, supports organizations in workforce planning, and helps educational institutions align programs with industry roles and progression.

Job Level | **Educational Level**

Business Intelligence & Strategy | **Data Stewardship** | **Data Engineering** | **Data Science** | **AI Engineering** | **Applied Data / AI Research** | **Data / AI Education**

C Level | **Doctoral / Post Doctoral**

Senior Director | **Chief [Business Function] Officer** | **Chief Data Officer** | **Chief Information Officer** | **Chief Analytics Officer** | **Chief Technology Officer** | **Chief Scientific Officer**

Director | **Business Analytics Director** | **Data Governance Officer** | **Chief Data Architect** | **Chief Data Scientist** | **Chief AI Engineer** | **Director of Research**

Senior Manager | **Business Analytics Manager** | **Data Governance Manager** | **Data Architect** | **Senior Data Scientist** | **Senior AI Engineer** | **Research Manager**

Manager | **Senior Business Intelligence Analyst** | **Data Quality Specialist** | **Senior Data Engineer** | **Data Scientist** | **AI Engineer** | **Senior Applied Data/AI Researcher**

Senior Professional / Supervisor | **Post Baccalaureate**

Professional | **Baccalaureate**

Business Intelligence Analyst | **Data Engineer** | **Machine Learning Engineer** | **Applied Data/AI Researcher**

Refer to the UNESCO AI Competency Framework for Teachers, www.unesco.org/en/articles/ai-competency-framework-teachers